

CO2-AT1- SCENARIOBASED ASSESSMENT

← Back

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing (10 Marks)
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- Enumerate three Collection interfaces suitable for this system.
- Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Map<String, Set<String>> studentCourseMap = new HashMap<>();
5
6         Set<String> aliceCourses = new HashSet<>();
7         aliceCourses.add("Java Programming");
8         aliceCourses.add("Data Structures");
9         aliceCourses.add("Database Management");
10
11        Set<String> bobCourses = new HashSet<>();
12        bobCourses.add("Python Programming");
13        bobCourses.add("Web Development");
14
15        studentCourseMap.put("Alice", aliceCourses);
16        studentCourseMap.put("Bob", bobCourses);
17        System.out.println("---- Student Registration Details ----");
18        Iterator<Map.Entry<String, Set<String>>> mapIterator = studentCourseMap.entrySet().iterator();
19
20        while (mapIterator.hasNext()) {
21            Map.Entry<String, Set<String>> entry = mapIterator.next();
22            String studentName = entry.getKey();
23            Set<String> courses = entry.getValue();
24
25            System.out.println("Student: " + studentName);
26            System.out.print("Enrolled Courses: ");
27
28            Iterator<String> courseIterator = courses.iterator();
29            while (courseIterator.hasNext()) {
30                System.out.print(courseIterator.next());
31                if (courseIterator.hasNext()) {
32                    System.out.print(", ");
33                }
34            }
35            System.out.println("\n-----");
36        }
37    }
38 }
```

Input

stdin Input

Output

Run

Save

CO2-AT1- SCENARIOBASED ASSESSMENT

← Back

QUESTIONS

Q1
Student Registration System
10 marks**Q2**
E-Commerce Cart System
10 marks**Q3**
Library Management System
10 marks**Q4**
Employee Data Processing (10 Marks)
10 marks**Q5**
Online Voting System
10 marks

5/5 answered

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- (a) List three Collection classes applicable for cart management.
(b) Represent how cart items are stored using a List-based structure.
(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.List;
4 public class Main {
5     public static void main(String[] args) {
6         List<String> cart = new ArrayList<>();
7         cart.add("Laptop");
8         cart.add("Wireless Mouse");
9         cart.add("Headphones");
10        cart.add("Keyboard");
11
12        System.out.println("Initial Cart Items added successfully.\n");
13        String itemToRemove = "Wireless Mouse";
14        Iterator<String> removeIterator = cart.iterator();
15
16        while (removeIterator.hasNext()) {
17            String item = removeIterator.next();
18            if (item.equalsIgnoreCase(itemToRemove)) {
19                removeIterator.remove();
20                System.out.println("Removed item: " + item + "\n");
21            }
22        }
23        System.out.println("---- Current Shopping Cart Items ----");
24        Iterator<String> displayIterator = cart.iterator();
25        int count = 1;
26
27        while (displayIterator.hasNext()) {
28            System.out.println(count + ". " + displayIterator.next());
29            count++;
30        }
31    }
32 }
```

Run

Save

Input

stdin input

Output

CO2-AT1- SCENARIOBASED ASSESSMENT
[← Back](#)
QUESTIONS
Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing (70 Marks)
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- Name three Map implementations in Java.
- Design a schema-like representation using Map for storing ISBN and book details.
- Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

Your Code

```
1 import java.util.HashMap;
2 import java.util.Map;
3 public class Main {
4     public static void main(String[] args) {
5         Map<String, String> bookMap = new HashMap<>();
6         bookMap.put("978-0134685991", "Effective Java");
7         bookMap.put("978-0596009205", "Head First Java");
8         bookMap.put("978-0132350884", "Clean Code");
9         System.out.println("Books added successfully.\n");
10        String searchIsbn = "978-0132350884";
11        if (bookMap.containsKey(searchIsbn)) {
12            System.out.println("Retrieved Book [ISBN: " + searchIsbn + "]: " + bookMap.get(searchIsbn));
13        } else {
14            System.out.println("Book with ISBN " + searchIsbn + " not found.");
15        }
16
17        System.out.println("\n-----");
18        System.out.println("--- Library Book List ---");
19        for (Map.Entry<String, String> entry : bookMap.entrySet()) {
20            System.out.println("ISBN: " + entry.getKey() + " | Title: " + entry.getValue());
21        }
22    }
23 }
```

 Run

 Save

Input

Output

QUESTIONS

Q1
Student Registration System
10 marks**Q2**
E-Commerce Cart System
10 marks**Q3**
Library Management System
10 marks**Q4**
Employee Data Processing (10 Marks)
10 marks**Q5**
Online Voting System
10 marks

5/5 answered

Q4 Employee Data Processing (10 Marks)

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.List;
4 class Employee {
5     private int id;
6     private String name;
7     private double salary;
8     public Employee(int id, String name, double salary) {
9         this.id = id;
10        this.name = name;
11        this.salary = salary;
12    }
13    public int getId() {
14        return id;
15    }
16    public String getName() {
17        return name;
18    }
19    public double getSalary() {
20        return salary;
21    }
22    @Override
23    public String toString() {
24        return "ID: " + id + " | Name: " + name + " | Salary: $" + salary;
25    }
26 }
27 public class Main {
28     public static void main(String[] args) {
29         List<Employee> employeeList = new ArrayList<>();
30         employeeList.add(new Employee(101, "Alice", 60000.0));
31         employeeList.add(new Employee(102, "Bob", 45000.0));
32         employeeList.add(new Employee(103, "Charlie", 75000.0));
33         employeeList.add(new Employee(104, "Diana", 48000.0));
34         employeeList.add(new Employee(105, "Ethan", 52000.0));
35         System.out.println("--- Employees with Salary > 50,000 ---");
36         Iterator<Employee> iterator = employeeList.iterator();
37
38         while (iterator.hasNext()) {
39             Employee emp = iterator.next();
40             if (emp.getSalary() > 50000) {
41                 System.out.println(emp);
42             }
43         }
44     }
45 }
```

Input

stdin input

Output

CO2-AT1- SCENARIOBASED ASSESSMENT

← Back

QUESTIONS

Q1
Student Registration System
10 marks**Q2**
E-Commerce Cart System
10 marks**Q3**
Library Management System
10 marks**Q4**
Employee Data Processing (10 Marks)
10 marks**Q5**
Online Voting System
10 marks

5/5 answered

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
(b) Design a structure using Set and Map for voters and vote counting.
(c) Write a Java program using HashSet and HashMap to:
- Record votes
 - Prevent duplicate voting
 - Display results

Your Code

```
1 import java.util.HashMap;
2 import java.util.HashSet;
3 import java.util.Map;
4 import java.util.Set;
5 public class Main {
6     private static Set<String> votedVoters = new HashSet<>();
7     private static Map<String, Integer> voteCountMap = new HashMap<>();
8     public static void castVote(String voterId, String candidate) {
9         if (votedVoters.contains(voterId)) {
10             System.out.println("Vote rejected for Voter " + voterId + " ");
11             return;
12         }
13         votedVoters.add(voterId);
14         voteCountMap.put(candidate, voteCountMap.getOrDefault(candidate, 0) + 1);
15         System.out.println("Vote successfully cast by " + voterId + " ");
16     }
17     public static void displayResults() {
18         System.out.println("\n--- Final Voting Results ---");
19         for (Map.Entry<String, Integer> entry : voteCountMap.entrySet()) {
20             System.out.println("Candidate: " + entry.getKey() + " | Votes: " + entry.getValue());
21         }
22         System.out.println("Total Unique Voters Voted: " + votedVoters.size());
23     }
24     public static void main(String[] args) {
25         castVote("V101", "Alice");
26         castVote("V102", "Bob");
27         castVote("V103", "Alice");
28         castVote("V101", "Bob");
29         castVote("V104", "Alice");
30         castVote("V102", "Alice");
31         displayResults();
32     }
33 }
```

Input

stdin input

Output

Run

Save